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Experimental investigation of light pipe and Window glazing retrofitting for improving indoor thermal conditions and daylight availability in existing buildings targeted for green retrofitting

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Enhancing the indoor environmental quality of existing buildings is crucial for realizing energy-efficient and user-friendly built environments. Traditional structures are usually subject to excessive sunlight penetration through windows and a lack of adequate daylight, leading to greater reliance on artificial illumination and mechanical air conditioning. Passive retrofitting methods are a viable initiative for improving interior comfort without significant structural restructuring. This research presents an experimental study on two passive retrofitting methods: light pipe daylighting systems and retrofitting double-glazed windows with the goal of improving indoor thermal conditions and daylight availability in buildings. Selected experimental conditions were tested in a 3 m × 3 m × 3 m cube test chamber. Environmental parameters, such as indoor air temperature, relative humidity and illuminance were determined hourly between 09:00 and 16:00 h. The findings show that the indoor temperature dropped by 2 °C–3 °C with the retrofit of double-glazed windows. Additionally, the light pipe system increased indoor illumination by more than 520 lux on average compared to the usual 170 lux thus meeting the recommended indoor lighting standard. The daylighting retrofit also achieved approximately 62.5% reduction in artificial lighting energy consumption, while the glazing retrofit reduced indoor temperature by up to 3.2 °C, indicating lower cooling demand. The normalized TVCI was developed as an exploratory comparative index using the normalization of comfort scores with regard to indoor temperature, relative humidity, and illuminance as per the internationally accepted comfort ranges. The combination of light pipe and double-glazed windows gave the highest value of the normalized TVCI, showing the closest agreement with internationally recommended indoor environmental conditions in terms of thermal and daylighting comfort levels. It is clear from the results that the use

of the combined retrofit method improves indoor environment efficiency and can possibly help to rely less on artificial lighting.

KEYWORDS

green building retrofitting, indoor daylight availability, light pipe day lighting, normalized thermal visual comfort index, thermal comfort, window glazing

1 Introduction

Buildings are one of the greatest consumers of energy worldwide, accounting for around 30%–40% of the whole world's energy consumption as well as a notable portion of greenhouse gas emissions (Han et al., 2010; Omer, 2008). A major portion of this energy demand is linked with the maintenance of acceptable indoor environmental conditions such as thermal comfort, lighting and ventilation (Duraković, 2026; Ismail et al., 2026). In many conventional buildings, inefficient building design combined with the lack of passive strategies for environmental control results in excessive energy consumption for artificial lighting and mechanical cooling systems (Zhang et al., 2019; Wu et al., 2026). Consequently, enhancing the performance of existing buildings by energy efficient retrofitting strategies has become a critical research focus in sustainable building development (Baños et al., 2011; Biyik and Kahraman, 2019). Existing conventional buildings, especially in developing countries, have a poor indoor environmental quality with poor insulation, inefficient window design and limited use of natural daylight (El-Darwish and Gomaa, 2017; Ahn et al., 2016). These deficiencies result in high indoor temperatures and insufficient daylight levels which have a negative impact on occupant comfort and productivity (Sansaniwal et al., 2021; Yüksel et al., 2021). Additionally, excessive reliance on artificial lighting systems leads to higher electricity use and costs of operation (Sakhri et al., 2022; Ran et al., 2022). Therefore, passive retrofitting strategies that improve thermal and visual comfort without demanding any significant changes to the building structure are necessary to transform conventional building into the green-to-be building (Vanhoutteghem et al., 2015; Evola et al., 2017).

Thermal comfort describes the state of mind which conveys satisfaction with the thermal environment. It is affected by a number of environmental parameters such as air temperature, relative humidity, air velocity and mean radiant temperature (Baranova et al., 2017; Seppçnen and Seppänen, 2008). Maintaining thermal comfort within acceptable limits is critical to occupant health and productivity (Awbi, 2017; Laverge et al., 2011). According to international building comfort standards, indoor thermal comfort conditions are generally achieved when air temperature is between 22 °C and 27 °C and relative humidity is between 40%–60%. In warm climatic regions, over heating due to excessive solar heat gain through windows and building envelopes is often the reason that indoor temperatures exceed these limits, which may require the application of energy-intensive cooling systems (Mahdavi Adeli et al., 2020; Nowak et al., 2016).

Visual comfort is another crucial element of indoor environmental quality and is the comfort in which the occupants can perform visual tasks without suffering from discomfort or eyestrain (Tabadkani et al., 2021; Fernandes and Regnier, 2023). Proper levels of illumination increase productivity, decrease fatigue and increase occupant wellbeing. International lighting standards suggest illuminance levels of 300–500 lux for common residential and office settings (David Babarinde and Alibaba, 2018; Mesloub et al., 2023). However, many existing buildings simply do not reach these lighting levels using natural daylight, so a greater reliance on artificial lighting is used during daylight hours. Daylighting systems are an effective way to enhance the visual comfort of a space and at the same time, to save electricity (Aghajari and Chen, 2025; Ma'bdeh and Al-Khatatbeh, 2019).

Among various passive daylighting technologies, light pipe daylighting systems have attracted a lot of attention because of its capabilities to harvest and convey sunlight from roof surfaces into interior spaces (Kwong, 2020; Secchi et al., 2015). Light pipes are a series of highly reflective tubular structures, used to transmit daylight from a rooftop collector to an indoor diffuser to evenly distribute the light in a room. These types of systems are especially useful in deep plan buildings where traditional windows are ineffective for providing daylight to the interior spaces (Ikuzwe et al., 2020; Knoop, 2016). Another passive retrofitting strategy that works well is window improvement with double glazed glass systems. Conventional single-glazed windows admit large amounts of solar heat energy that raises indoor temperatures and contribute to thermal discomfort (Shin et al., 2012; Canziani et al., 2004). Double-glazed windows have two layers of glass between them, with an air space. Heat transfer through conduction and radiation is reduced. This configuration enhances the thermal insulation of the windows, and will help to keep the indoor temperatures stable, thus improving the thermal comfort (Zhang et al., 2002; Bhutani et al., 2024). Although a number of studies have been carried out in the fields of thermal comfort improvement and daylighting separately, most of the existing research have relied heavily on numerical simulations and computational models (Malet-Damour et al., 2019; Mushtaha et al., 2016). Experimental investigations of the joint effect of passive daylighting systems and window retrofitting in terms of

Abbreviations: T, Indoor air temperature (°C); T_o, Outdoor air temperature (°C); T_i, Indoor air temperature (°C); RH, Relative humidity (%); I, Indoor illuminance (lux); E_i, Indoor illuminance used in daylight factor calculation (lux); E_o, Outdoor illuminance (lux); DF, Daylight factor (%); TVCI, Thermal Visual Comfort Index; Q, Rate of heat transfer (W); U, Overall heat transfer coefficient (W/m^2K); A, Surface area of glazing (m^2); I_o, Incident light intensity (lux); I_t, Transmitted light intensity (lux); R, Reflectivity of light pipe internal surface; n, Number of internal reflections; a,b,c,d, Empirical regression coefficients; R^2, Coefficient of determination; S_T, Temperature Comfort Score; S_{RH}, Humidity Comfort Score; S_I, Illuminance Comfort Score; w_T, Weighting coefficient for temperature comfort score; w_{RH}, Weighting coefficient for relative humidity co; w_I, Weighting coefficient for illuminance comfort score; TVCI, Thermal Visual Comfort Index; HVAC, Heating, Ventilation and Air Conditioning; SHGC, Solar Heat Gain Coefficient; U-value, Overall heat transfer coefficient; RCC, Reinforced Cement Concrete.

indoor environmental conditions are relatively scarce. Furthermore, most previous studies assess the thermal and visual comfort as separately, without taking into account their combined effects on overall indoor comfort conditions (Martins Mogo de Nadal, 2005; Beltrán et al., 1997). To overcome these limitations, this study is presented, which is an experimental investigation of passive retrofitting strategy in order to enhance both thermal comfort and indoor daylight availability at the same time. A controlled experimental setup representing a typical residential space was created and 2 types of retrofitting strategies were established and tested under actual environmental conditions: light pipe daylighting systems and double glazed window retrofits (Azad and Rakshit, 2018; Šikula et al., 2014). In addition to the classical thermal and visual comfort analysis, this research presents a normalized Thermal and Visual Comfort Index (TVCI) which combines thermal and visual parameters into one comfort analysis measure. Apart from enhancing occupant comfort levels, passive retrofitting techniques have the ability to significantly cut down energy needs for building operations by reducing reliance on electric lighting and mechanical cooling devices. In hot climates, cooling and lighting account for a significant portion of household energy usage. Thus, measuring the savings achieved through the deployment of daylighting technology and thermal glass is an important step in the assessment of sustainable building retrofits.

The key objective of this experimental research work lies in evaluating the efficiency of the passive retrofitting approaches towards improving indoor environmental quality as illustrated in Figure 1 below. In particular, the experiment analyzes the effect of double-glazed windows on indoor thermal environment along with the influence of light pipe daylighting systems on the indoor daylight conditions. Furthermore, the joint impact of the two retrofitting methods under consideration is explored in order to evaluate the impact on indoor environmental conditions as well as to compare them. Finally, the integrated indoor environmental performance is evaluated using the experimental normalized Thermal Visual Comfort Index (TVCI), which has been proposed by the researchers in the current research as a tool for evaluating joint thermal comfort and indoor daylight availability conditions. In addition, the research estimates the possibility of reducing artificial lighting due to increased daylight availability.

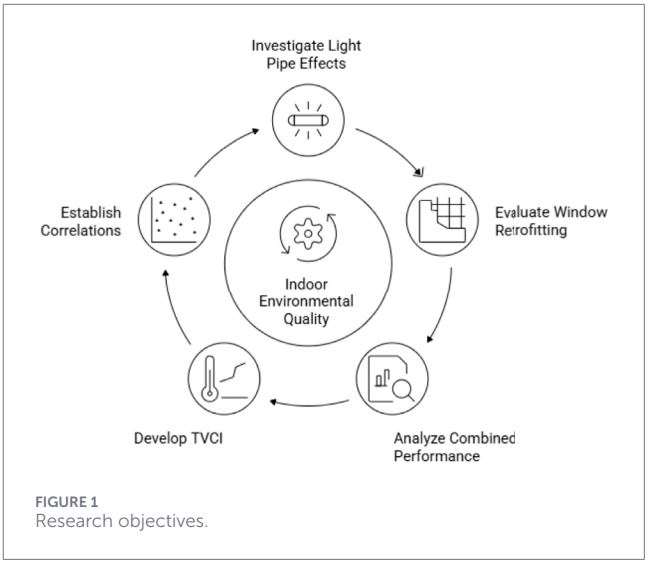


FIGURE 1
Research objectives.

2 Literature review

The increasing need for sustainable building performance has led to a large number of studies on energy efficiency, indoor environmental quality, and passive retrofit measures. Among these, thermal comfort, visual comfort and renewable energy integration are considered as key parameters to perform better building and decrease energy consumption for operation. This section is a review of previous studies that are related to thermal comfort, daylighting systems, window retrofitting technologies, and integrated indoor comfort evaluation in buildings.

2.1 Energy efficiency in existing buildings

Existing conventional buildings comprise a very large percentage of the building stock worldwide and are generally known for their poor energy performance, resulting from ineffective building envelopes, ineffective insulation, and the lack of passive design features. Retrofitting these buildings with energy efficient technologies is a practical way of achieving sustainable development (Hanjani et al., 2024; Sharma et al., 2018). Previous studies have highlighted the fact that energy efficient retrofitting measures like insulation upgrades, better glazing systems, daylighting technologies and renewable energy integration can offer significant reduction in building energy consumption. According to various building energy assessments, the lighting and cooling load combined accounts for almost 60%–70% of the total electricity consumption in residential buildings especially in warm climatic regions. Therefore, a better use of the daylight and a lower solar heat gain are relevant strategies to minimize energy demands (Goharian and Mahdavinejad, 2020; Jenkins and Muneer, 2004). Research on the development of green building has illustrated the necessity to make use of energy efficient retro-fitting techniques that could affect the quality of the indoor environment without being too expensive. Retrofitting approaches that do not necessitate significant structural changes are particularly useful when it comes to adapting conventional buildings to become green-to-be buildings, since they can be accomplished with little construction complexity and cost (Omishore et al., 2019; Carter, 2002).

2.2 Thermal comfort studies in buildings

Thermal comfort is a very important parameter in the overall indoor environmental quality of buildings. It is affected by several environmental factors and personal factors such as air temperature, relative humidity, air velocity, clothing insulation, and metabolic rate (Jenkins et al., 2005; Hansen, 2026). Thermal comfort at acceptable limits is required to maintain occupant health, productivity and satisfaction. Several researchers have studied thermal comfort conditions in residential and commercial buildings through the use of experimental and simulation-based methods. Studies conducted in hot and dry climates have shown that passive cooling techniques such as shading devices, natural ventilation and high performance building envelopes can reduce

the indoor temperatures significantly and improve the comfort of the occupants. In particular, shading devices and better window designs have been found to reduce the solar heat gain and to prevent overheating in highly glazed buildings. Research in the field of ventilation strategies has shown that improved ventilation systems can be used to increase thermal comfort at lower energy cost. For instance, some studies on the performance of residential buildings with varying ventilation configurations have shown energy savings of up to 29% while maintaining acceptable indoor temperature levels (Gil-Martín et al., 2014; Paroncini et al., 2007). These results show that with proper design of building envelopes and ventilation strategies, high levels of thermal comfort can be achieved without the need for high levels of mechanical cooling systems. In addition to passive design strategies, a number of studies have been conducted to determine the role of renewable energy technologies in maintaining comfortable indoor conditions. Net-zero energy building concepts are often a combination of the use of photovoltaic systems, passive cooling strategies and energy-efficient HVAC systems to achieve improved thermal performance. However, thermal comfort problems can not be solved by the integration of renewable energy alone, unless it is combined with good building envelope design (Wu et al., 2024; Jenkins and Muneer, 2003; Fguiri et al., 2025).

2.3 Daylighting systems and indoor daylight availability

Visual comfort is another crucial element of the internal environmental quality and is determined mostly by availability of daylight, light uniformity, glare control, and color rendering. Proper daylighting design has the benefit of fewer reliance on artificial lighting systems as well as better occupant wellbeing. Daylighting strategies are often comprised of windows, skylights, clerestories, and tubular daylighting devices (Teixeira et al., 2020). Among such, light pipe daylighting systems have emerged as a promising solution in delivering natural light to interior spaces, located at considerable distance from external walls. Light pipes are made of highly reflective tubes which capture the sun's rays at the roof surface and transmit the rays into interior spaces by multiple internal reflections. Studies on daylighting technologies have reported daylight pipe systems have the potential of reducing artificial lighting energy consumption by approximately 20%–30% in buildings. These systems are especially useful in deep plan buildings where the conventional windows cannot achieve enough daylight penetration (Li and Wu, 2025). Experimental investigations have shown that reflective light pipes with optimal diameters and reflective coatings can be used to achieve indoor illumination levels up to more than 500 lux, which are adequate for recommended lighting levels for residential and office spaces. Despite their benefits, daylighting systems should be well designed to avoid problems such as glare and excessive heat gain (Shaik et al., 2022; Mainini et al., 2015). The combination of daylighting technologies with suitable designs of building envelope is thus required in order to achieve balanced indoor environmental conditions (Lee and Song, 2023; Vanhoutteghem et al., 2015).

2.4 Window design and glazing technologies

Windows play an important role in controlling the transfer of heat and daylight into buildings. Conventional single-glazed windows admit a large amount of solar radiation into the internal spaces resulting in elevated indoor temperatures and thermal discomfort (Ran et al., 2022). In addition, inefficient window designs contribute to increasing cooling loads in warm climates (Ge et al., 2026; Ganesh et al., 2021). Double-glazed window has been widely studied as an effective solution for improving thermal performance of building envelope performance. These windows are made up of two panes of glass with an air space in between to provide an insulating layer so that heat is not easily transferred from the interior to the exterior or *vice versa*. The performance of glazing systems in terms of thermal effects is often measured by parameters such as U-value (a measure of the rate of heat passing through the window) and Solar Heat Gain Coefficient (SHGC), which is a measure of the proportion of the solar radiation passing through the glazing. Experimental and simulation-based researches have revealed that double-glazed windows can considerably freeze the indoor heat gain and enhance the thermal comfort condition. In the case of residential buildings, optimized glazing systems have been shown to cut cooling energy demand with sufficient daylighting levels (Nowak et al., 2016). However, too much glazing area or inadequate orientation may still cause overheating and thus the importance of careful design and combination with other forms of passive strategies (Asere and Blumberga, 2018; Underhill et al., 2020).

2.5 Combined thermal and indoor daylight availability evaluation

Although many studies have examined thermal comfort comfort and indoor daylight availability individually, there are relatively low numbers of studies that have examined the combined effect of both dimensions on indoor environmental quality. In actual building settings the occupants not only experience thermal but also visual conditions and thus an integrated approach of evaluation is required to correctly evaluate the level of indoor comfort. Previous studies have suggested different comfort indices for mixing the environmental parameters to assess indoor conditions (Derbez et al., 2014; Kumar et al., 2016). However, most of these indices give a main attention to thermal comfort parameters such as temperature and humidity, while visual comfort parameters are often treated separately. Recent research has stressed the importance of integrated comfort models that address thermal and visual environmental parameters together and at the same time. The development of integrated comfort indices is quite important for the evaluation of passive retrofit strategies, where improvements in daylight availability may also affect thermal conditions due to solar radiation. Therefore, a well-rounded evaluation framework of comfort, that captures thermal and visual parameters, is necessary in order to properly judge the performance of passive building retrofits (Ortiz et al., 2020; Wells et al., 2015).

2.6 Research gap identification

Based on the literature review, some of the important research gaps have been identified. Most of the existing studies on daylighting systems and glazing technologies rely mostly on simulation-based approaches and have a relatively small amount of experimental validation in real building environments. Furthermore, previous research has usually investigated thermal comfort and indoor daylight availability as two different aspects, without giving sufficient attention to how the two aspects interact and influence the quality of their combined effect. There is also a lack of experimental studies to assess the integrated performance of light pipe daylighting systems in combination with window glazing retrofits for improving indoor comfort conditions. In addition, there are quite few studies that have suggested comprehensive comfort indices in which both thermal and visual environmental parameters are considered at the same time. In order to overcome these limitations, the current research study conducts a controlled experimental study of light pipe and window glazing retrofitting strategies using a full-scale test chamber. Moreover, it presents a normalized Thermal Visual Comfort Index (TVCI) for evaluation of integrated conditions of indoor comfort based on experimentally measured environmental parameters (Dascalaki and Sermpetzoglou, 2011; Silva et al., 2022; Kanika et al., 2016; Vasilakopoulou et al., 2016).

While the existing literature has mainly focused on either daylighting techniques or glazing systems individually or by applying simulation methods, the present research attempts to examine both techniques together under the same measurement framework using a full-scale laboratory testing system. Moreover, a novel normalized Thermal Visual Comfort Index (TVCI) is introduced for comparative indoor environment assessment.

3 Experimental methodology

This section explains the experimental setup, instrumentation, retrofitting configurations, measurement procedures and analytical methods for the evaluation of the performance of passive retrofit strategies. The experimental investigation was designed to mimic a typical residential indoor environment and quantify the effect of light pipe daylighting systems and double-glazed window retrofits on indoor thermal conditions and daylight availability (Persily and Emmerich, 2012; Bashir et al., 2026; Shawon et al., 2026). The experiments were carried out at Greater Noida, Uttar Pradesh, India, having a humid subtropical climate. The experiments were conducted in April-May under summer conditions of clear sky.

3.1 Experimental test room design

A controlled chamber of an experimental room was designed to mimic a typical room in a dwelling schematically shown in Figure 2. The dimensions of the chamber were 3 m × 3 m x 3 m, which is a small residential living space with details mentioned in Table 1.

The south-facing window orientation has been chosen as south-facing building facades receive maximum solar radiation during the daytime hours and hence can be used for the analysis of heat gain and daylight penetration. The internal surface of the room was covered by light-colored paint (reflectance approx. 0.65) to simulate typical residential internal conditions; this will enhance the daylight distribution.

3.2 Passive retrofitting systems

Two passive retrofitting systems were implemented to improve indoor environmental conditions.

3.2.1 Light pipe daylighting system

A tubular daylighting device (light pipe) was installed on the roof of the experimental chamber in order to harvest and convey natural daylight into the room schematically shown in the Figure 3. The light pipe system consists of three main parts that together work to effectively transmit natural daylight into indoor spaces. The collector dome is on the roof of the house, and it captures incident solar radiation and transfers it into the system. This light is then passed through a high reflecting tube where numerous internal reflections are used to ensure that daylight travels through with a minimum amount of loss. Finally, the diffuser panel distributes the transmitted light in a homogeneous way within the room and thus contributes to a higher indoor daylight availability and a reduction of glare. The internal reflectivity of the pipe is high and this plays a very important role in keeping the illumination efficiency of the entire transmission process. A pipe of 300 mm in diameter and 1.2 m long formed the light pipe daylighting system. The inside of the pipe was highly reflective with a 95% reflectivity level. The acrylic dome was used as an outdoor collector, and the prismatic diffuser was responsible for distributing the light indoors.

3.2.2 Double-glazed window retrofit

The existing single glazed window has been replaced with a double glazed window unit to reduce solar heat gain and enhance thermal insulation schematically shown in Figure 4. The air gap between the two layers of glass serves as a thermal barrier against the transfer of heat through conduction helping to decrease fluctuations of indoor temperature. The double glazed window was made up of two sheets of glass, each measuring 6 mm in thickness, with a gap of 12 mm filled with air between them. This window design had a U-value of 2.7 W/m^2K, showing that there was less heat loss, while its SHGC value of 0.63 assisted in minimizing solar heat gain inside.

3.3 Experimental configurations

Four experimental configurations were tested in order to compare the performance of the retrofitting strategies as mentioned in the Table 2. The experimental study was carried out using four separate types of passive retrofits to analyze their effects on indoor environmental parameters. The first case considered the situation where no passive retrofit had been made, thus acting as a control case. Case B included the use of double glazed windows for improving thermal performance of the building envelope. Case C consisted of the light pipe daylighting system for improving the level of natural lighting indoors. Case D included both the light pipe daylighting system and double glazed windows. Each configuration was tested under similar outdoor environmental conditions to ensure that there was consistency in comparing data. Environmental

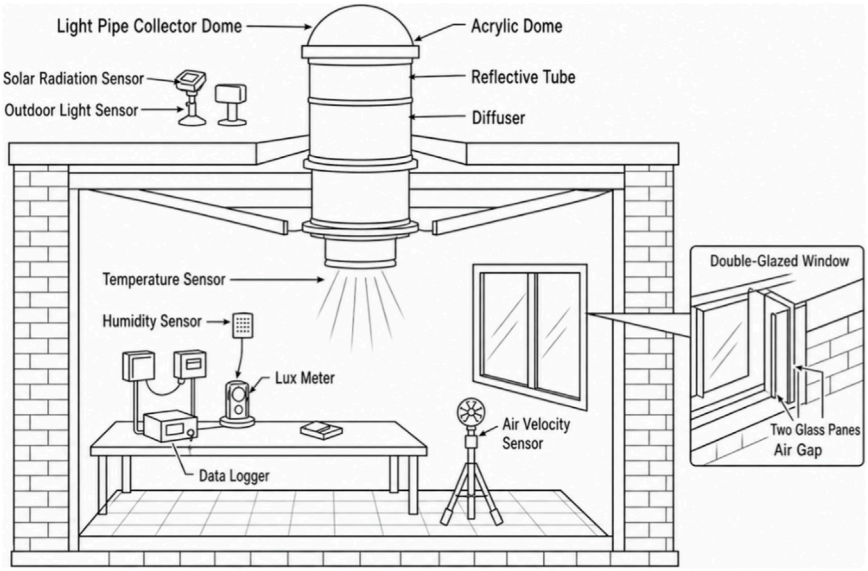


FIGURE 2
Schematic Diagram of sample test environment.

TABLE 1 Experimental room design parameters and specifications.

Parameter	Specification
Room dimensions	3 m × 3 m × 3 m
Wall material	230 mm brick masonry
Internal per	12 mm cement plaster
Roof structure	Reinforced concrete slab
Floor material	Ceramic tile
Window size	1.2 m × 1.2 m
Window orientation	South facing

parameters were tested with calibrated instruments with proper levels of accuracy to ensure reliable data collection.

All instruments were calibrated in advance of the experimental campaign to reduce the measurement uncertainty. In order to obtain representative indoor environmental measurements, the sensors were placed strategically at the level of the occupant working plane, at 1.1 m above the floor. Measurements were made at three different spots in the space: in the center of the room, near the window, and on the opposite side of the room, in order to ensure a comprehensive evaluation of the spatial variations in the indoor conditions. The illuminance sensor however was at a height of 0.8 m to match typical desk or task levels commonly used in indoor lighting studies. In addition, outdoor illuminance sensors were installed on the roof to simultaneously capture incident daylight conditions and the indoor and outdoor lighting performance can be accurately compared.

3.4 Experimental procedure

The experiments were conducted during clear sky days to ensure stable solar radiation conditions. At each measurement interval key environmental parameters were systematically recorded in order to assess the indoor conditions on a comprehensive basis. These included outdoor temperature to record ambient climatic conditions, indoor temperature to assess thermal performance, relative humidity to determine moisture levels in the space, and indoor illuminance to assess the performance of sorting daylight and lighting distribution within the room. The experiments carried out at 1-h intervals during the testing time period, which ranged from 09:00 to 16:00 h. For each configuration of retrofitted lighting systems, 5 days were utilized in order to make sure the accurate results were collected. The outdoor illuminance during the testing periods varied between 60,000 and 90,000 lux. The mean results of the observation days were used to represent the environmental conditions for each retrofit configuration. Daylight factor (DF) is a common metric that is used to assess daylight availability inside buildings. It is the ratio of illuminance of an indoor space to illuminance of the outdoor space under identical sky conditions. Because these tests were performed using real clear sky and not standard CIE overcast sky, the figures for daylight factors are presented as experimental daylight ratios for comparison purposes in relation to the current study.

$$DF = \frac{E_i}{E_o} \times 100$$

Higher daylight factor values represent higher daylight penetration. For residential spaces, the range of daylight factor values, in general, is between 0.5% and 2%. Clear sky was chosen for conditions in which the amount of clouds was less than 10% and variations in outside illuminance was ±10%. No one was inside the test room. There were no heat-producing devices in the room. The testing room was kept closed during tests.

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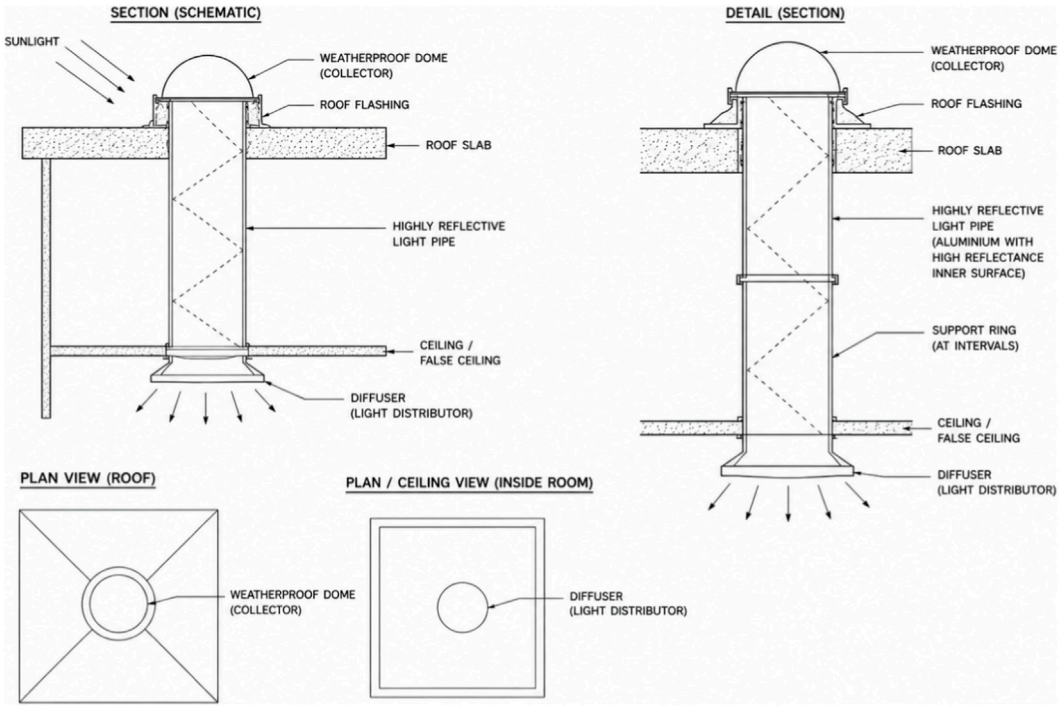


FIGURE 3
Schematic diagram of Solar lite pipe.

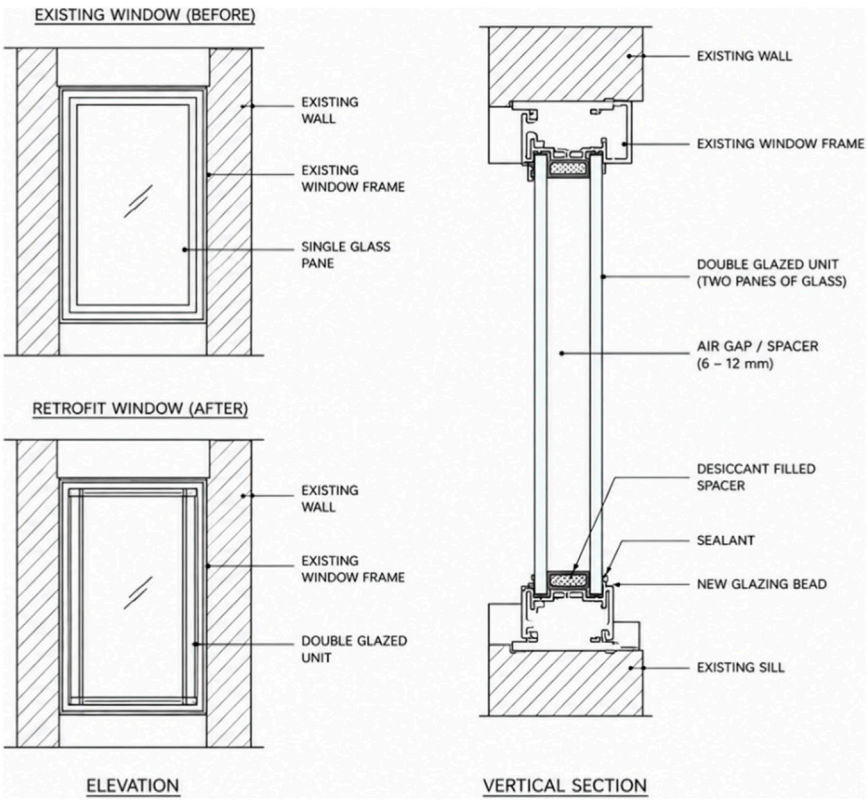


FIGURE 4
Illustration of double-glazed window retrofit.

TABLE 2 Measurement instruments and accuracy specifications.

Parameter	Instrument	Accuracy
Indoor air temperature	Temperature data logger	±0.5 °C
Relative humidity	Digital hygrometer	±3%
Air velocity	Hot-wire anemometer	±0.03 m/s
Indoor illuminance	Lux meter	±3%
Outdoor illuminance	Digital lux meter	±3%
Energy consumption	Digital energy meter	±1%

3.5 Normalized Thermal Visual Comfort Index (TVCI) using the Entropy Weight Method

The aim of creating a Normalized Thermal Visual Comfort Index (TVCI) was for an integrative comparison of indoor environmental performance in the studied retrofit alternatives. Different from existing indices like PMV/PPD and adaptive comfort model, the created TVCI does not measure or estimate thermal sensation of human beings. On the contrary, it is designed to serve as a comparative environmental performance index that combines normalized scores of comfort levels for temperature, humidity, and illuminance inside buildings. The three environmental factors are first converted into normalized scores of their comfort levels before integration into one composite index.

The proposed empirical relation is given as:

$$TVCI = w_T S_T + w_{RH} S_{RH} + w_I S_I$$

The weighting coefficients satisfy:

$$w_T + w_{RH} + w_I = 1$$

ensuring that the normalized TVCI varies between 0 and 1, where higher values indicate closer agreement with internationally recommended indoor environmental comfort ranges.

In the case of the temperature comfort score, the maximum value can be obtained when the actual measured indoor temperature is inside the recommended comfort zone and then gradually declines when the actual measured indoor temperature diverges from the recommended comfort zone. In a similar way, the relative humidity comfort score will have the highest value for the recommended comfort zone of 40%–60% and the illuminance comfort score will have the maximum value in the recommended indoor daylighting zone of 300–500 lux. As such, high TVCI values imply that the indoor environment is close to meeting the international standards of comfort with regards to thermal and lighting aspects. Thus, it can only serve as an indicator of environmental performance as opposed to thermal sensations of occupants. The proposed TVCI is intended solely for comparative evaluation of the investigated retrofit configurations and should not be interpreted as a replacement for PMV/PPD or adaptive comfort models. The choice of the Entropy Weight Method was based on the fact that this method evaluates the criteria weights objectively depending on the information contained in the experimental data.

3.5.1 Normalization of indoor environmental parameters

Prior to computing the normalized Thermal Visual Comfort Index, each of the environmental parameters was converted to a normalized comfort value between 0 and 1. The value of 1 shows that the environmental conditions are in the recommended range for indoor comfort conditions, whereas the values near 0 show that the environmental condition is away from the recommended range. In order to calculate the normalized Thermal Visual Comfort Index (TVCI), all the measured parameters of the indoor environment were converted to a normalized comfort level that could be evaluated on a scale from 0 to 1, depending on how close they were to internationally recognized recommendations for indoor environments. Specifically, regarding the indoor air temperature, the range of thermal comfort temperatures recommended by ASHRAE Standard 55 and ISO 7730, which was 22 °C–27 °C, was assumed to be an optimum one in terms of thermal comfort, and any value of the temperature within this range was awarded the maximum comfort level. In case of indoor relative humidity, the values within the recommended range of 40%–60% were supposed to be providing optimal humidity levels inside a building, thus awarding the highest comfort score to these values, and the values outside of this range received lower scores depending on how far they were from the recommended ones. Concerning the indoor daylight availability, the recommended illuminance values of 300–500 lux, according to EN 12464–1 recommendations, provided the optimal indoor illuminance conditions and received the maximum comfort score. Therefore, the TVCI developed in this study measures the degree of compliance between the measured values of indoor environmental parameters and internationally recognized comfort recommendations. The weighting factors were calculated objectively via the use of the Entropy Weight Method (EWM), where the criteria weights are obtained based on the variation of the experimental data points as opposed to any subjective reasoning.

3.5.1.1 Normalization

$$P_{ij} = \frac{x_{ij}}{\sum_{i=1}^{m} x_{ij}}$$

where:

x_{ij} = measured value of parameter j for observation i;
p_{ij} = normalized proportion of parameter j corresponding to observation i;
m = total number of observations;

3.5.1.2 Entropy

$$E_j = -k \sum_{i=1}^{m} P_{ij} \ln\left(P_{ij}\right)$$

where:

j = environmental parameter (temperature, relative humidity, or illuminance).
E_j = entropy value of parameter j;
k = entropy constant

$$k = \frac{1}{\ln(m)}$$

3.5.1.3 Degree of diversification

$$D_j = 1 - E_j$$

where:

D_j = degree of diversification (information utility) of parameter j;
E_j = entropy value of parameter j.

A larger value of D_j indicates greater variability in the experimental data and consequently greater information content.

3.5.1.4 Weight

$$w_j = \frac{D_j}{\sum D_j}$$

where:

w_j = entropy-derived weighting coefficient of parameter j;
D_j = degree of diversification of parameter j;
n = total number of evaluated environmental parameters (three in this study: temperature, relative humidity, and illuminance).

The calculated weighting coefficients satisfy

$$\sum_{j=1}^{n} w_j = 1$$

The normalization process involves the conversion of all the measured environmental variables to dimensionless comfort scores between 0 and 1, with a score of 1 meaning full adherence to the indoor comfort ranges recommended and a score close to 0 meaning deviation from the recommended values. Subsequently, the Entropy Weight Method (EWM) is used objectively to determine the weight of each of the normalized variables using the variability in the experimental data. Finally, the normalized TVCI score is calculated by combining the three normalized comfort scores using the determined weights.

3.6 Data analysis

The collected experimental data were analyzed using statistical and regression methods to find relations between the most important factors affecting the parameters of the environment and the indoor comfort conditions. The analysis involved a comparison of temperature and illuminance trends in various retrofit configurations to determine their performance. In addition to this, daylight factor calculations were performed for several experimental cases to quantify the daylight availability. The study was also carried out through the computation of the normalized Thermal Visual Comfort Index (TVCI) to assess the integrated comfort conditions by combining both the thermal and the visual aspects.

4 Experimental results and data analysis

This part introduces the experimental observations obtained from the four test configurations and analyses the effect of the passive retrofitting strategies on the indoor thermal comfort and indoor daylight availability. The measured parameters are indoor air temperature, relative humidity and indoor illuminance which were recorded every 1 hour between 09:00 and 16:00 under clear sky conditions. The experimental analysis is concentrated on the identification of the influence of the light pipe system and double glazed window retrofit on the indoor environmental performance and the evaluation of the improvement using normalized Thermal Visual Comfort Index (TVCI). In order to make sure that all the four retrofit designs could be compared on an equal basis, it is necessary to note that all the experiments were carried out with similar outdoor conditions. Outdoor temperatures varied in the limits of 31.2 °C–38.2 °C, whereas outdoor illuminances varied in the limits of 60,000 to 90,000 lux. Moreover, any deviations among various experimental setups did not exceed ±5%. Consequently, differences in indoor climates could be explained by the application of retrofitting techniques.

4.1 Outdoor environmental conditions

During the experimental period, the outdoor environmental conditions were relatively stable and typical of the warm climatic conditions as enlisted in Table 3. The measured outdoor temperature had 31.2 °C–38.2 °C, and outdoor illuminance values achieved about 80000 lux when it had solar intensity at peak around midday. These outdoor conditions provide the boundary environment for evaluating indoor performance under different retrofit configurations.

4.2 Indoor thermal performance

Indoor temperature variances were examined in all the experimental cases to assess the effects of glazing retrofits and daylighting systems depicted by Figure 5. The baseline condition illustrates a continuous rise in the indoor temperature from the solar heat gain through the building envelope and window surfaces. The maximum indoor temperature recorded was 34.6 °C, which is significantly higher than recommended thermal comfort limits. The double glazed window performed significantly to reduce solar heat gain and so a reduction in temperature of around 2 °C–3 °C was achieved compared to the baseline condition. The insulating air space between the glass panes is reducing conductive heat transfer and thermal transmission to enhance the thermal performance of the window system. The light pipe system brings natural daylight in the room but does not change the internal

TABLE 3 Outdoor environmental conditions during experimental period.

Time	Outdoor temperature (°C)	Outdoor illuminance (lux)
09:00	31.2	62000
10:00	33.5	70000
11:00	35.7	76000
12:00	37.4	80000
13:00	38.2	82000
14:00	37.8	79000
15:00	36.4	72000
16:00	34.9	65000

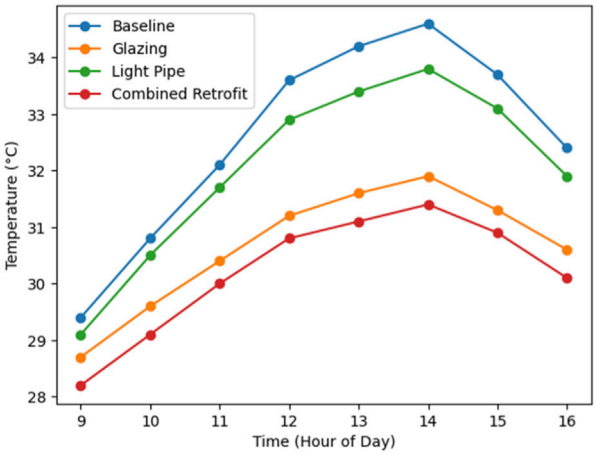


FIGURE 5
Indoor temperature variation for different retrofit cases.

temperature of the room to any major extent. However, there may be slight variations in temperature due to the solar radiation passing into the collector dome. The combined configuration recorded the most favorable values for indoor temperature, the maximum temperature was 31.4 °C representing a reduction of about 3.2 °C compared to the baseline condition. Table 4 shows comparative hourly variation of indoor temperature under different passive retrofitting configurations.

4.3 Daylight availability analysis

Indoor illuminance levels were tested for the effectiveness of the daylighting systems. The baseline configuration shows insufficient daylight levels, with illuminance values significantly below the recommended 300–500 lux indoor lighting range. The double-glazed window retrofit slightly improved indoor illuminance levels by allowing controlled daylight transmission while reducing excessive solar heat gain. The light pipe system dramatically improved the indoor daylight availability. Peak illuminance values were 600 lux, which is sufficient for recommended indoor lighting. The combined retrofit configuration provided the greatest levels of indoor illumination, which provided comfortable illumination levels throughout the observation period as depicted in Figure 6. Table 5 shows comparative hourly variation of indoor illuminance under different passive retrofitting configurations. It is important to mention that the visual evaluation made during this study focused mainly on indoor illuminance levels. Parameters such as glare, luminance distribution, uniformity of natural light, contrast, and occupant visual comfort were not evaluated within this study; therefore, the improvement can be understood only in terms of natural daylight availability.

4.4 Daylight factor evaluation

The daylight factor was calculated by using the measurement data made at noon (12:00). At this time the illuminance of the inside was measured 610 lux and the outside illuminance of the same was 80,000 lux. These values were used to assess the availability of daylight in the space to the outside lighting conditions.

$$DF = \frac{610}{80000} \times 100$$

$$DF = 0.76\%$$

A computed result of 0.76% shows that there is moderate penetration of sunlight in the experimental room under the present clear sky condition. It should be noted that the value obtained above is a daylight ratio which is based on the experimental clear sky condition and thus should not be confused with the conventional daylight factor which is calculated under the standard overcast sky condition according to the CIE standard.

4.5 Normalized Thermal Visual Comfort Index (TVCI)

To evaluate integrated comfort performance, the normalized TVCI was calculated using the proposed empirical relation:

$$TVCI = w_T S_T + w_{RH} S_{RH} + w_I S_I$$

TABLE 4 Comparative hourly variation of indoor temperature under different passive retrofitting configurations.

Time	Baseline condition (°C)	Double-glazed window retrofit (°C)	Light pipe daylighting system (°C)	Combined light pipe + glazing retrofit (°C)
09:00	29.4	28.7	29.1	28.2
10:00	30.8	29.6	30.5	29.1
11:00	32.1	30.4	31.7	30
12:00	33.6	31.2	32.9	30.8
13:00	34.2	31.6	33.4	31.1
14:00	34.6	31.9	33.8	31.4
15:00	33.7	31.3	33.1	30.9
16:00	32.4	30.6	31.9	30.1

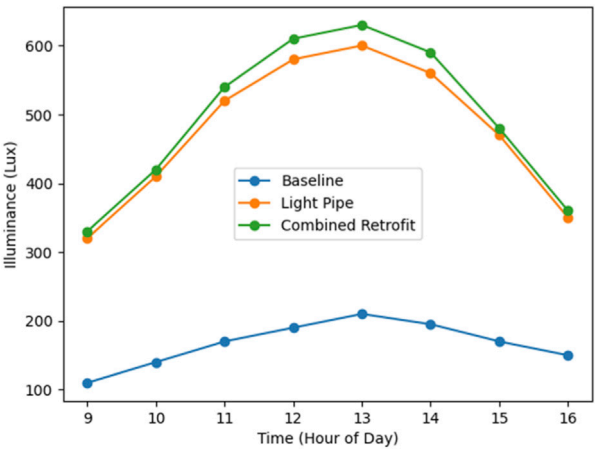


FIGURE 6
Indoor illuminance variation for different test cases.

Calculation for combined retrofit at 12:00 for T = 30.8 °C, RH = 44%, I = 610 lux; Temperature comfort score, $S_T = 0.55$; Relative humidity comfort score, $S_{RH} = 1.00$; Illuminance comfort score, $S_I = 1$. Entropy-derived weighting coefficients are calculated and shown in Table 6.

Now,

$$TVCI = 0.41S_T + 0.18S_{RH} + 0.41S_I$$

From the derived entropy weights, it is apparent that both indoor air temperature and illuminance contribute almost equally to the environmental evaluation process, while relative humidity contributes less variably amongst the retrofit configurations analyzed and thus has a smaller weight factor associated with it. It is worth noting that the calculated weight factors are data driven. Increased normalized TVCI values suggest that the indoor environmental conditions being assessed are more closely related to international recommendations for thermal and daylighting comfort ranges. Therefore, the developed TVCI may be considered an exploratory comparative tool for assessing indoor environmental performance but not necessarily a general thermal comfort index model.

$$TVCI = 0.41(0.55) + 0.18\,(1.00) + 0.41\,(1.00)$$

$$TVCI = 0.8155$$

$$TVCI = 81.55\%$$

Thus, the combined retrofit configuration achieved a normalized TVCI of approximately 81.55%, indicating that the measured indoor environmental conditions were closest to the recommended thermal and daylighting comfort ranges among the investigated retrofit configurations.

4.6 Comparative retrofit performance

The combined retrofit configuration offers the best compromise in both thermal comfort and indoor daylight availability conditions proving the effectiveness of combining daylighting systems and improved window insulation. The fact that there is a decrease in indoor temperature and increase in daylight availability shows significant opportunities for savings on energy consumption in such building designs. In addition, the integration of these two systems decreased indoor temperature by 3 °C–4 °C, while still ensuring adequate lighting inside the building above the indoor lighting standards. Table 7 shows comparative performance of passive retrofitting strategies.

5 Exploratory analysis of indoor environmental parameters

In order to further understand the relationship between indoor environmental parameters and overall comfort conditions, empirical models were developed based on the experimental data set obtained from the 4 retrofit configurations. The objective of this analysis is to establish mathematical correlations which may

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TABLE 5 Comparative hourly variation of indoor illuminance under different passive retrofitting Configurations.

Time	Baseline condition (lux)	Double-glazed window retrofit (lux)	Light pipe daylighting system (lux)	Combined light pipe + glazing retrofit (lux)
09:00	110	125	320	330
10:00	140	155	410	420
11:00	170	185	520	540
12:00	190	205	580	610
13:00	210	225	600	630
14:00	195	210	560	590
15:00	170	180	470	480
16:00	150	165	350	360

TABLE 6 Entropy-derived weighting coefficients.

Parameter	Entropy	Diversification	Weight
Temperature comfort score	0.912	0.088	0.41
Relative humidity comfort score	0.961	0.039	0.18
Illuminance comfort score	0.875	0.125	0.41

be used to investigate empirical relationships among the measured indoor environmental parameters and evaluate the comparative indoor environmental performance of the investigated retrofit configurations.

5.1 Dataset preparation for empirical analysis

The experimental data were the hourly measurements of the indoor temperature, relative humidity, and illuminance data collected from the various retrofit configurations. For regression modelling, representative average values for the observation period were used. Table 8 shows normalized Thermal Visual Comfort Index (TVCI) comparison for different retrofit cases. Figure 7 shows the TVCI vs. Retrofit Configuration. The normalized TVCI values were calculated using the empirical relation introduced earlier:

$$TVCI = w_T S_T + w_{RH} S_{RH} + w_I S_I$$

5.2 Empirical relationship interpretation

The regression models developed in this study offer a number of important insights into the indoor environmental performance. The results show that the illuminance level inside the space is strongly affected by the daylighting systems (light pipes) to increase the availability of natural light inside the space. Additionally, it is demonstrated that the use of double-glazed windows substantially lowers heat transfer, which contributes to lower indoor temperatures and thermal comfort. Under the suggested formulation, increasing values of TVCI mean that the indoor environmental parameters are increasingly consistent with those recommended on an international basis regarding the thermal comfort and indoor daylight availability zones. This is achieved by combining the normalized indices of comfort in terms of temperature, relative humidity, and illuminance levels to provide an evaluation based on comparison rather than high numerical values of the environmental parameters. On the whole, the findings show that passive retrofitting strategies can dramatically enhance the indoor environment quality without the need for mechanical systems. Furthermore, the empirical equations resulting from this analysis can be used as practical tools to predict comfort conditions inside retrofitted buildings and can be used by architects and engineers to design energy-efficient and sustainable building solutions.

5.3 Measurement uncertainty and error analysis

Experimental measurements always contain some uncertainties because of the accuracy of the measuring instruments, the variation of the measuring environment, and the variation in the measuring procedures. Therefore, uncertainty analysis was performed in order to determine the reliability of the collected environmental data.

5.3.1 Instrument uncertainty

The measurement uncertainties related to the instruments that were used in the experiment are summarized below in Table 9.

5.3.2 Combined measurement uncertainty

The combined uncertainty on calculated parameters including normalized Thermal Visual Comfort Index (TVCI) can be estimated by root-sum-square method:

$$U = \sqrt{(U_T)^2 + (U_{RH})^2 + (U_I)^2}$$

TABLE 7 Comparative performance of passive retrofitting strategies.

Retrofit strategy	Temperature reduction	Illuminance improvement
Glazing	2 °C–3 °C	Minor
Light pipe	Minimal	300%–400%
Combined retrofit	3 °C–4 °C	400%

TABLE 8 Comparative evaluation of normalized thermal visual comfort index (TVCI) for different retrofit Configurations.

Case	Temperature (°C)	RH (%)	Illuminance (lux)	ST	SRH	SI	TVCI
Baseline	33.1	44	167	0.3	1	0.35	55
Glazing	30.7	44	177	0.5	1	0.37	62.3
Light pipe	32	44	476	0.38	1	0.95	77.7
Combined retrofit	30.2	44	495	0.55	1	1	81.55

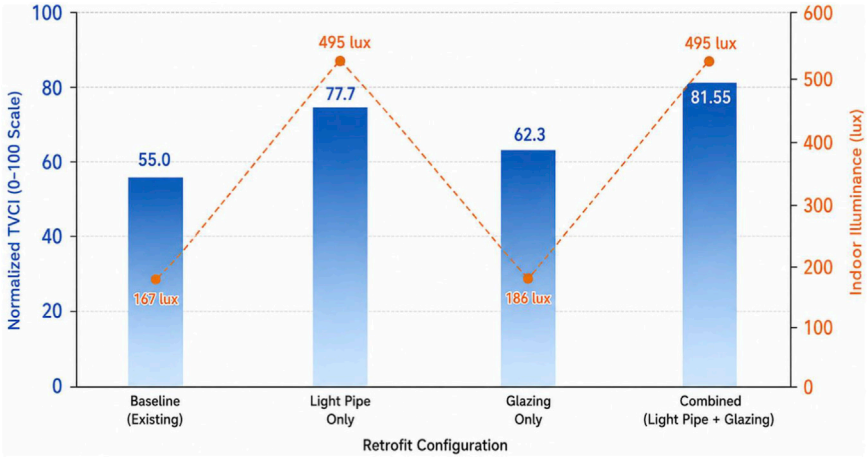


FIGURE 7
Normalized TVCI vs. Retrofit Configuration.

TABLE 9 Measurement instruments, accuracy, and uncertainty specifications.

Parameter	Instrument	Accuracy	Uncertainty
Temperature	Temperature data logger	±0.5 °C	±0.5 °C
Relative humidity	Digital hygrometer	±3%	±3%
Illuminance	Lux meter	±3%	±3%
Air velocity	Hot wire anemometer	±0.03 m/s	±0.03 m/s

Where:

1. U = combined uncertainty, U_T = uncertainty in temperature measurement,
2. U_{RH} = uncertainty in relative humidity measurement,
3. U_I = uncertainty in illuminance measurement.

Substituting the experimental uncertainties:

$$U = \sqrt{(0.5)^2 + (3)^2 + (3)^2}$$

$$U = \sqrt{0.25 + 9 + 9}$$

$$U = \sqrt{18.25}$$

$$U = 4.27$$

Thus, the upper limit of experimental uncertainty of combined environmental measurements is about ±4.3% which is acceptable in the experiments of buildings environment.

5.4 Energy saving analysis

One of the major benefits of daylighting systems is the decrease in energy used by artificial lighting systems.

The lighting energy saving percentage is calculated using:

$$Energy\ Saving(\%) = \frac{E_b - E_r}{E_b} \times 100$$

Where:

E_b = baseline lighting energy consumption, E_r = energy consumption after retrofit.

Assuming that the room supplied by an originally done 20 watt LED lamp for 8 h a day.

Baseline energy consumption:

$$E_b = 20 \times 8$$

$$E_b = 160\,Wh/day$$

After installation of the light pipe, artificial lighting was required for only 3 h per day.

$$E_r = 20 \times 3$$

$$E_r = 60\,Wh/day$$

Energy saving:

$$Energy\ Saving = \frac{160 - 60}{160} \times 100$$

$$Energy\ Saving = 62.5\%$$

Therefore, from the scenario-based assessment, it can be seen that there is a possible reduction of about 62.5% in lighting energy usage. It needs to be pointed out that the current energy saving assessment is made under the assumption about operating time of the lighting system (8 h/day before the improvement and 3 h/day after the retrofit). Thus, the value stated above should be seen only as the possible estimate of energy savings.

6 Discussion

The experimental investigation performed in the present study can provide valuable information on the effectiveness of passive retrofitting strategies for the improvement of indoor environmental quality in existing buildings. The combined consideration of thermal comfort, indoor daylight availability and the proposed normalized Thermal Visual Comfort Index (TVCI) shows the potential to combine daylighting technologies with increased insulation of the windows to improve occupant comfort while reducing energy usage.

6.1 Influence of window glazing on thermal performance

The experimental results show that after replacing the conventional single-pane window and the double-pane window system, the thermal performance of the experimental chamber was greatly improved. The measured decrease of indoor temperature varied from 2 to 3 °C, which can be attributed to the insulating air layer between two glass panes which mitigate the conductive heat transfer. In reference to double glazing windows, the U-value is less in comparison to single glazing; hence the heat transfer across the window surface is less. This increase in thermal resistance is responsible for lower temperatures inside the house and improved thermal comfort conditions. The results of the current study are in agreement with the previous researches that reported about the improvement of thermal insulation and the reduction of cooling load in the buildings by double glazed window. Studies on energy efficient building envelopes have shown that optimized glazing systems can play an important role in reducing the solar heat gain and ensure still sufficient daylight penetration. Thermal evaluation was mainly done using temperatures and relative humidities. The mean radiant temperature, clothing insulation, metabolic rate, and occupant adaptation were not considered in this study.

6.2 Impact of light pipe system on daylight availability

The light pipe daylighting system showed a significant improvement in indoor illuminance levels as compared to baseline configuration. The experimental results indicated that the indoor illumination improved from average 170 lux at baseline case to about 520 lux at the installation of light pipe system. This improvement is achieved because of the high internal reflectivity of the light pipe tube which enables daylight to be transmitted efficiently from the rooftop collector to the interior diffuser. The reflective tube reduces losses of light and combines the daylight evenly inside the room. Since the internal reflectivity of the pipe was estimated to be about 95%, the reduction in light intensity during transmission was found to be minimal and adequate daylight was able to reach interior spaces. The resulting indoor illumination levels were more than 500 lux at peak daylight hours, which is at the level recommended in living and office spaces. This means that light pipe systems can effectively minimise the need for artificial lighting during the day hours. In this particular study, the measurement of indoor daylight availability was confined to illuminance alone. Other factors, such as glare, luminance distribution, day-lighting uniformity, contrast ratio, and occupant perception, were not considered.

6.3 Performance of combined retrofitting strategy

This combination of light pipe and double-glazed window retrofit achieved the best indoor environmental condition of all of the configurations. This integrated approach resulted in a reduction of indoor temperature of about 3.2 °C and at the same time, increased indoor illumination to more than 600 lux during maximum daylight hours. The observed synergy of these two strategies can be explained by their complementary roles: while the double glaze window is an efficient strategy to minimize solar heat gain and thus reduce the indoor temperature, the light pipe system promotes efficient daylight penetration to provide adequate indoor light, without notably adding to the thermal load. Consequently, the combined retrofit configuration balances the indoor thermal conditions and daylight availability, which improves the overall indoor environmental quality. Experimental results show that the use of the suggested passive retrofitting technique makes an additional contribution to improvements in indoor environment quality in addition to operational energy savings. Given the marked rise in natural sunlight, it would seem feasible to cut down on lighting needs. Based on the estimated usage schedule, a scenario estimate showed that it was possible to reduce energy expenditure on lighting by about 62.5%. Likewise, since there is a drop in interior temperature levels, it could be reasonable to assume less energy spent on cooling.

6.4 Evaluation of normalized Thermal–Visual Comfort Index (TVCI)

The combined retrofit produced the highest normalized TVCI because the measured indoor environmental parameters were collectively closest to the recommended thermal comfort and indoor daylight availability ranges. The suggested normalized TVCI includes the summation of the individual thermal comfort and indoor daylight availability measures and not just the physical parameters. The implication is that the index measures how close the observed conditions within an indoor environment are to the comfort limits and not the magnitude of the physical values like temperature, humidity, and illuminance. The newly revised normalized TVCI scores have been evaluated against the exploratory regression-based estimates to examine the consistency of the empirical relationships observed within the experimental dataset. From Table 9, it can be observed that the regression-based TVCI scores were quite close to the experimentally calculated normalized TVCI scores, with absolute differences of 1.03% and 1.45%. Given that the newly revised TVCI score is based on normalized comfort scores obtained from the recommended indoor environmental values, the regression analysis has been performed just as an exploratory evaluation of data relationships.

6.5 Comparison with previous studies

The results of this research are consistent with the previous study of passive daylighting and window retrofitting technologies. Several studies have indicated that daylighting systems like light pipes can achieve substantial improvements in the quality of indoor illumination at the price of reduced lighting energy use. Similarly, double glazed window systems have been widely known to be able to improve the thermal insulation and reduce the cooling loads in buildings. However, most of the previous studies have considered either thermal comfort and indoor daylight availability solely, whereas in the present study, both parameters are considered at the same time using an integrated comfort index. The experimental approach followed in the present research also, overcomes the limitation of simulation-based studies, by providing the experimental validation of the passive retrofit performance in the real world.

6.6 Practical implications for green-to-Be buildings

The results of this study show how passive retrofitting strategies can play an important role in improving the indoor environmental conditions in existing buildings without the need for major structural modifications. The combination of light pipe daylighting systems and double glazed windows offers a number of practical benefits; the reduction in indoor temperature and improved thermal comfort, improved daylight penetration resulting in increased comfort from light; and reduced dependence on artificial light. These improvements also add to a possible drop in overall electricity consumption. Collectively, the results suggest that the proposed retrofitting strategy can be an effective and feasible strategy to transform conventional buildings into green and energy-efficient buildings in order to promote sustainable building development and encourage energy conservation.

6.7 Study limitations

However, some limitations should also be mentioned in relation to the findings obtained within the framework of this research work. First of all, this laboratory research was performed using only one 3 m × 3 m × 3 m test chamber under clear-sky conditions, and thus these results can be viewed as proof-of-concept outcomes and should not be applied to buildings constructed in other climatic regions. Moreover, thermal evaluation of the tested lighting fixture was carried out considering only two parameters–temperature and relative humidity–whereas no information about mean radiant temperature, clothing insulation, metabolic rate, and occupant adaptation was obtained during testing. At the same time, visual evaluation was performed according to the illuminance level only, and there was no assessment of such parameters as glare, luminance distribution, daylight uniformity, contrast ratio, and perception of occupants. Moreover, normalized Thermal Visual Comfort Index should be considered as an exploratory comparative index. Also, the energy saving values presented in this paper are estimates obtained based on lighting operation scenarios and are not experimentally determined.

7 Conclusion

The current experiment was conducted to explore the viability of passive retrofits as means for enhancing indoor environmental quality in existing buildings with the help of 3 m × 3 m × 3 m test chamber. Two types of retrofits–light pipe daylighting systems and double-glazed window retrofits were evaluated in four configurations, including baseline configuration, glazing retrofit, light pipe installation, and combination of both retrofits. In this

regard, the present research sought to examine indoor thermal conditions, daylight availability, and overall indoor environmental performance through the suggested normalised Thermal Visual Comfort Index (TVCI). Data concerning indoor air temperature, relative humidity, and illuminance level were recorded using calibrated tools under normal outdoor conditions. Besides, daylight factor analysis and empirical modeling was used to examine empirical relationships among the measured indoor environmental parameters. As a result, installation of double glazing reduced indoor air temperature by about 2 °C–3 °C, limiting heat transfer and heat gains from solar energy, hence, providing stable indoor thermal conditions. The light pipe technology significantly enhanced the illuminance level inside the building raising its value from almost 170 lux–520 lux with up to 600 lux during peak daylight period. The combination of both retrofits demonstrated excellent results, allowing obtaining up to 3.2 °C indoor temperature reductions and maintaining illuminance levels above 500 lux, leading to an estimated potential lighting energy saving of approximately 62.5% under the assumed lighting operation schedule. The suggested normalized TVCI includes the summation of the individual thermal comfort and indoor daylight availability measures and not just the physical parameters. The implication is that the index measures how close the observed conditions within an indoor environment are to the comfort limits and not the magnitude of the physical values like temperature, humidity, and illuminance.

8 Future scope

Although the current study shows the effectiveness of passive retrofitting strategies in terms of improving indoor environmental conditions, there are various opportunities for conducting further research and development. Future investigations can be directed towards extended studies of the performance of light pipe systems and glazing technologies under different climatic conditions can be conducted through long term performances in extended experimental studies over different seasons. Another important direction involves integration of passive retrofitting technologies with renewable energy technologies (e.g., photovoltaic panels), which might further improve overall building energy performance and sustainability. Additionally, research can investigate advanced daylight control systems, such as automated shading devices and smart daylight sensors, to dynamically optimize indoor lighting conditions. Further studies should also include occupant centered comfort studies in the form of surveys and physiological measurements to validate and refine the proposed normalized Thermal Visual Comfort Index (TVCI). Expanding the scope of the experiments to larger building environments, such as multi-room or multi-zone buildings, would be useful for assessing the ability to scale-up and practically apply passive retrofitting strategies for real-world applications. Moreover, the development of more advanced comfort models is needed, on which the normalized Thermal Visual Comfort Index can be improved with other environmental parameters such as air velocity, mean radiant temperature and glare index. Advancements in these areas of research will help to create more comprehensive building performance evaluation methodologies and can help to drive the further adoption of sustainable retrofit strategies in existing buildings. Future studies need to validate the proposed normalized TVCI through occupant survey in conjunction with existing thermal comfort models such as PMV/PPD and Adaptive Comfort.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

VB: Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review and editing. PB: Project administration, Supervision, Writing – review and editing. SS: Project administration, Supervision, Writing – review and editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

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